

CSE464Project1

Features per commits

1. Feature 1

- `parseGraph(filepath)`
- `toString()`
- `outputGraph(filepath)`

2. Feature 2

- `addNode(label)`
- `addNodes(labels[])`
- duplicate node handling

3. Feature 3

- `addEdge(srcLabel, dstLabel)`
- duplicate edge handling

4. Feature 4

- `outputDOTGraph(path)`
- `outputGraphics(path, format)`

CSE464Project2

Features per commits

<https://github.com/eavila12/CSE4642026eavila12/commit/1a3b0a67ac619276f7432e7fae8fccf0501d391c>

- `removeNode(String label)`
- `removeNodes(String[] labels)`
- `removeEdge(String srcLabel, String dstLabel)`

BFS Branch

<https://github.com/eavila12/CSE4642026eavila12/commit/ada613cc19561bd898f98050b3008e98d436d59b>

- `Path GraphSearch(Node src, Node dst)`

DFS Branch

<https://github.com/eavila12/CSE4642026eavila12/commit/7e8d62a1785a6ca53ad7d6de386ef0286c02c8be>

- Path GraphSearch(Node src, Node dst)

Final Merged Main

<https://github.com/eavila12/CSE4642026eavila12/commit/e95d63493e4a6c1aef00df59d3273d9eaa070fd4> <https://github.com/eavila12/CSE4642026eavila12/commit/0e7647ca371394b58be44b70982543747ffdb550>

- Path GraphSearch(Node src, Node dst, Algorithm algo)

CI

GitHub Actions workflow

link: <https://github.com/eavila12/CSE4642026eavila12/actions> Successful CI run on main: <https://github.com/eavila12/CSE4642026eavila12/actions/runs/2397285183> Successful CI run on

bfs: <https://github.com/eavila12/CSE4642026eavila12/actions/runs/23972586780> Successful CI run on

dfs <https://github.com/eavila12/CSE4642026eavila12/actions/runs/23968327462> To test the code run the following command to build

`mvn clean test package`

Once the build is successful run the following command

`mkdir -p out mvn -q exec:java -Dexec.mainClass=edu.asu.cse464.dot.App -Dexec.args="./input.dot ./out"`

the example input is

`digraph G { a -> b; b -> c; a; "node x" -> c; }`

The plain text out put is

```
Nodes (4): [a, b, c, node x]
Edges (3):
  a -> b
  b -> c
  node x -> c
```

and the image should be

